

ON SEPARATING SEMIGROUPS OF $(M-2)$ -CURVES

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ABSTRACT. A separating morphism from a real algebraic curve C to $\mathbb{P}^1$ is a real morphism f satisfying $f^{-1}(\mathbb{RP}^1) = \mathbb{R}C$. If $C_1, \dots, C_r$ are the connected components of $\mathbb{R}C$, then every separating morphism determines a vector $d(f) = (d_1(f), \dots, d_r(f))$, where $d_i(f)$ is the degree of the restriction of f to C_i . The set of all such vectors is called the separating semigroup of C .

In the present paper we study separating semigroups of $(M-2)$ -curves, that is, real genus g curves whose real locus has $g-1$ connected components.

We prove that the separating semigroup is equal to $\mathbb{N}^{g-1}$ on a non-empty open subset of the Teichmüller space of such curves. In the remaining cases, we establish inclusions that generalize the known low-genus results.

§1. Introduction

By a *real algebraic curve* C we mean a complex algebraic curve endowed with an antiholomorphic involution $\text{conj}: C \rightarrow C$, called a *real structure*. The set of real points of C , denoted by $\mathbb{R}C$, is the fixed-point set of conj . All curves considered here are projective, smooth, and irreducible.

A real curve C is called *separating* or *dividing* if $C \setminus \mathbb{R}C$ is disconnected. In this case $C \setminus \mathbb{R}C$ consists of two connected components which are interchanged by an antiholomorphic involution. The boundary orientation induced on $\mathbb{R}C$ by one of the halves of $C \setminus \mathbb{R}C$ is called a *complex orientation*.

A classical result of Ahlfors [Ahl50] states that a real curve C is separating if and only if it admits a *separating morphism*, that is, a real morphism $f: C \rightarrow \mathbb{P}^1$ such that $f^{-1}(\mathbb{RP}^1) = \mathbb{R}C$.

Every separating morphism defines a covering map $f|_{\mathbb{R}C}: \mathbb{R}C \rightarrow \mathbb{RP}^1$. Fix a numbering of the connected components $C_1, \dots, C_r$ of $\mathbb{R}C$. Denoting by $d_i(f)$ the degree of the restriction of f to C_i , we can assign to each separating morphism f the tuple of positive integers $d(f) = (d_1(f), \dots, d_r(f))$. Kummer and Shaw defined the *separating semigroup* of C as the set of all such tuples:

$$\text{Sep}(C) := \{d(f) \mid f: C \rightarrow \mathbb{P}^1 \text{ is a separating morphism}\}.$$

It is shown in [KS20, Prop. 2.1] that $\text{Sep}(C)$ is indeed a semigroup. We denote $\mathbb{N} := \{n \in \mathbb{Z} \mid n \geq 1\}$ and $\mathbb{N}_0 := \{n \in \mathbb{Z} \mid n \geq 0\}$.

By the celebrated Harnack's inequality, we have $b_0(\mathbb{R}C) \leq g+1$ for every real curve C of genus g . A real curve C is called an M -curve if equality holds, that is, if $b_0(\mathbb{R}C) = g+1$. If $b_0(\mathbb{R}C) = g+1-i$, then C is called an $(M-i)$ -curve. Kummer and Shaw [KS20, Theorem 1.7] proved that for an M -curve C of genus g we have $\text{Sep}(C) = \mathbb{N}^{g+1}$. By Klein's congruence, for a separating $(M-i)$ -curve C we have $b_0(\mathbb{R}C) \equiv g+1 \pmod{2}$, so i must be even. Consequently, after M -curves, the next natural case to consider is that of $(M-2)$ -curves. The present paper is devoted to describing their separating semigroups.

Date: September 8, 2026.
Grant.

Brief review. The separating semigroups have already been computed for $(M-2)$ -curves of small genus. If $g = 3$, then $\text{Sep}(C) = (2, 1) + \mathbb{N}_0^2$ or $\text{Sep}(C) = \{(1, 1)\} \cup (\mathbb{N}_{\geq 2} \times \mathbb{N}_{\geq 2})$ (see [Ore19]). If $g = 4$, then, as shown in [Ore25], $\text{Sep}(C)$ is either $\mathbb{N}^3$, $(2, 1, 1) + \mathbb{N}_0^3$, or $\{(1, 1, 1)\} \cup ((2, 1, 1) + \mathbb{N}_0^3)$. If C is a plane $(M-2)$ -quintic (in which case $g = 6$), then $\text{Sep}(C) = (2, 1, 1, 1, 1) + \mathbb{N}_0^5$ (see [MO25]). The detailed review of low-genus cases is given in Appendix B.

The description of the separating semigroup depends crucially on the minimal degree of a separating morphism. We denote this value by $\text{sepgon}(C)$ and call it the *separating gonality* of C . It follows from results of A. Gabard [Gab06] that for an $(M-2)$ -curve C of genus g , we have $g-1 \leq \text{sepgon}(C) \leq g$, so only two cases are possible: $\text{sepgon}(C) = g-1$ or $\text{sepgon}(C) = g$.

In [Man26], M. Manzaroli studies the connection between separating morphisms and totally real pencils of curves of degree $d-3$ on plane separating $(M-2)$ -curves of degree d . In addition, that paper contains partial results on the separating semigroups of $(M-2)$ -curves of arbitrary genus. More precisely, it is shown that $(3, \dots, 3) + \mathbb{N}_0^{g-1} \subseteq \text{Sep}(C)$ if $\text{sepgon}(C) = g-1$, and $(4, 2, \dots, 2) + \mathbb{N}_0^{g-1} \subseteq \text{Sep}(C)$ if $\text{sepgon}(C) = g$.

Besides separating gonality, an important role is played by the notion of *special type* introduced by Coppens [Cop14]. To avoid overloading the introduction, we recall the definition and the basic properties of this notion in §2.2.

Main results. For curves not of special type, we obtain a complete answer: as in the case of M -curves, every vector of positive degrees is realizable by some separating morphism.

Theorem 3.7. *Let C be a separating $(M-2)$ -curve of genus g . Assume that $g \geq 5$ and C is not of special type. Then*

$$\text{Sep}(C) = \mathbb{N}^{g-1}.$$

It turns out that, for separating $(M-2)$ -curves, separating gonality and the special-type condition are closely related to theta characteristics. One of the main results of Coppens [Cop14] states that if C is a genus $g \geq 3$ $(M-2)$ -curve of special type with $\text{sepgon}(C) = g-1$, then every separating pencil of degree $g-1$ is a theta characteristic. We prove the converse for $g \geq 5$: if C is not of special type, then it admits a separating pencil of degree $g-1$ which is not a theta characteristic. Thus, among separating $(M-2)$ -curves with separating gonality $g-1$, the special type condition is characterized by the half-canonicity of all minimal separating pencils (for a detailed formulation see Corollary 3.6). This characterization is one of the main ingredients in the proof of Theorem 3.7.

Combining Theorem 3.7 with the results of Coppens [Cop14], we conclude that the separating semigroup equals $\mathbb{N}^{g-1}$ on a non-empty open subset of the Teichmüller space of separating $(M-2)$ -curves of genus g (Corollary 3.8).

In the two remaining cases, we obtain the following inclusions.

Theorem 3.1. *Let C be a separating $(M-2)$ -curve of genus $g \geq 4$.*

- (a) *Assume that $\text{sepgon}(C) = g$ and let f be a separating morphism of degree g . Number the connected components of $\mathbb{R}C$ so that $d_1(f) = 2$. Then*

$$(2, 1, \dots, 1) + \mathbb{N}_0^{g-1} \subseteq \text{Sep}(C).$$

- (b) *Assume that $\text{sepgon}(C) = g-1$ and C is of special type with a special component C_1 . Then*

$$\{(1, \dots, 1)\} \cup ((2, 1, \dots, 1) + \mathbb{N}_0^{g-1}) \subseteq \text{Sep}(C).$$

It is worth noting that equality holds in all known cases listed above (see Appendix B). This leads to the following conjecture.

Conjecture 1.1. *The inclusions in Theorem 3.1 are in fact equalities.*

Remark 1.2. We would also like to emphasize that, even in the case of plane quintics, proving the reverse inclusion in the description of $\text{Sep}(C)$, that is, ruling out all other degree partitions, requires a delicate argument that relies heavily on the fact that the curve is plane; see [MO25, Lemma 4.3]. This suggests that the general case may be substantially more difficult.

Organization of the paper. The paper is organized as follows. In §2 we first recall the necessary facts about separating divisors, including the infinitesimal version of Abel-Jacobi theorem which allows to construct separating divisors (Lemma 2.4) and the theorem of Mikhalkin and Orevkov on the extendability of separating divisors of degrees g and $g - 1$ (Theorem 2.5). We then review Coppens's notion of special type and its relation to separating gonality. The section concludes with the Brill-Noether-theoretic estimates used to control the locus of pencils with base points. In §3 we prove the main results. We first establish the inclusions of Theorem 3.1. We then turn to curves not of special type: the key step is to construct a separating pencil of degree $g - 1$ which is not a theta characteristic, allowing us to apply the extendability theorem and prove Theorem 3.7. The convex-geometric lemma underlying this construction is proved in Appendix A. In Appendix B, we determine the special type and separating gonality of separating $(M - 2)$ -curves of low genus.

Acknowledgements. The author is deeply grateful to Grigory Mikhalkin for fruitful discussions.

§2. Preliminaries

In this section, C denotes a smooth projective real algebraic curve of genus g with $b_0(\mathbb{R}C) = l$.

As usual, for a divisor D on C , we write $h^i(D) := \dim H^i(C, \mathcal{O}_C(D))$. For a base-point-free complete linear system $|D|$ of rank r , we denote the corresponding morphism $C \rightarrow \mathbb{P}^r$ by $f_{|D|}$.

A divisor D is called *real* if $\text{conj}(D) = D$. A real effective divisor is called *totally real* if its support is contained in $\mathbb{R}C$.

The following elementary observation will be very useful for us.

Proposition 2.1 ([Hui01, Corollary 2.2]). *Let $k < l$ be a natural number, and let $p_1, \dots, p_k \in \mathbb{R}C$ be points such that no two of them lie on the same connected component of $\mathbb{R}C$. Then the divisor $D = p_1 + \dots + p_k$ is rigid, that is, $h^0(D) = 1$.*

Lemma 2.2. *Let $f: C \rightarrow \mathbb{P}^1$ be a separating morphism of degree $l = b_0(\mathbb{R}C)$, and let P be a real fiber of f . Then $h^0(P) = 2$.*

Proof. Since f is separating and $\deg f = l$, a real fiber of f has the form

$$P = p_1 + \dots + p_l, \quad p_i \in C_i,$$

where $C_1, \dots, C_l$ are the connected components of $\mathbb{R}C$. For every i , Proposition 2.1 gives $h^0(P - p_i) = 1$, and hence $h^0(P) \leq h^0(P - p_i) + 1 = 2$. On the other hand, since P is a fiber of f , we have $h^0(P) \geq 2$. $\square$

For a real algebraic curve C , the Picard variety $\text{Pic}^d(C)$ is naturally endowed with an antiholomorphic involution. Its real locus $\mathbb{R}\text{Pic}^d(C)$ is diffeomorphic to a disjoint union of 2^{l-1} real g -dimensional tori.

A divisor class $[D] \in \text{Pic}^{g-1}(C)$ is called a *theta characteristic* if $2D \sim K_C$. By definition, any two theta characteristics differ by an element of the 2-torsion subgroup of the Jacobian of C . Since this subgroup is isomorphic to $(\mathbb{Z}/2\mathbb{Z})^{2g}$, there are exactly 2^{2g} pairwise inequivalent theta characteristics on C .

§2.1. Separating divisors. For the rest of this subsection, we assume that C is separating and write $C_1, \dots, C_l$ for the connected components of $\mathbb{R}C$.

For our purposes, it is sometimes more convenient to work with divisors in the corresponding linear systems rather than with separating morphisms themselves.

We call a divisor $P = p_1 + \dots + p_n$ on C *separating* if it is a fiber of a separating morphism; that is, if there exist a separating morphism $f: C \rightarrow \mathbb{P}^1$ and a point $t \in \mathbb{R}\mathbb{P}^1$ such that $P = f^{-1}(t)$. In this case, we write $d(P) := d(f)$; equivalently, $d_i(P) = \deg(P|_{C_i})$.

Lemma 2.3. *Let $C_1, \dots, C_l$ be the connected components of $\mathbb{R}C$, and let $D = p_1 + \dots + p_l$ with $p_i \in C_i$. Suppose that $|D|$ is base-point-free and $h^0(D) \geq 2$. Then D is separating.*

Proof. Since $|D|$ is base-point-free, there exists a real divisor $E \in |D|$, disjoint from D . For every $i = 1, \dots, l$ we have $\deg(D|_{C_i}) \equiv \deg(E|_{C_i})$ (see e.g. [Hui03, Corollary 2.4]). Thus E has odd degree on every connected component of $\mathbb{R}C$. Therefore, $\deg E = b_0(\mathbb{R}C) = l$ implies that E has exactly one point on each connected component of $\mathbb{R}C$. Hence divisors D and E interlace: every connected component of $\mathbb{R}C \setminus E$ contains exactly one point of D , and vice versa. It follows from [KS20, Lemma 2.10] that D is separating. $\square$

Another useful tool for working with separating divisors is the Abel–Jacobi theorem. Let $\omega_1, \dots, \omega_g$ be a real basis of $H^0(C, K_C)$.

Lemma 2.4 (Lemma 1 in [Mag26], cf. Lemma 3.2 in [Ore19]). *Let $p_1, \dots, p_n \in \mathbb{R}C$ be pairwise distinct points. Then the divisor $P = p_1 + \dots + p_n$ is separating if and only if there exist tangent vectors $v_i \in T_{p_i}(\mathbb{R}C)$, positive with respect to one of the two complex orientations of $\mathbb{R}C$, such that*

$$(AJ) \quad \sum_{i=1}^n \omega_k(v_i) = 0 \quad \text{for all } k = 1, \dots, g.$$

Note that condition (AJ) can be reformulated as follows. The tuple $v = (v_1, \dots, v_n)$ defines a tangent vector to $\text{Sym}^n(C)$ at P . Let $\varphi: \text{Sym}^n(C) \rightarrow \text{Pic}^n(C)$ denote the Abel–Jacobi map. Then condition (AJ) is precisely the statement that $v \in \text{Ker}(d_P\varphi)$.

Following [MO26], we call an element $d \in \text{Sep}(C)$ *extendable* if

$$d + \mathbb{N}_0^l \subseteq \text{Sep}(C).$$

We call a separating divisor P on C *extendable* if the corresponding element $d(P) \in \text{Sep}(C)$ is extendable.

As noted above, Gabard’s bound implies that every separating $(M - 2)$ -curve of genus g admits a separating morphism of degree either $g - 1$ or g . The following recent theorem of Mikhalkin and Orevkov provides sufficient conditions under which separating divisors of degree g or $g - 1$ are extendable.

Theorem 2.5 ([MO26]). *Let C be a separating real curve of genus g with $b_0(\mathbb{R}C) = l$.*

- (a) *Let $P = p_1 + \dots + p_g$ be a separating divisor on C with $h^1(P) = 1$. Then $d(P) + \mathbb{N}_0^l \subseteq \text{Sep}(C)$.*
- (b) *Assume that C is not hyperelliptic. Let $Q = q_1 + \dots + q_{g-1}$ be a separating divisor with $h^0(Q) = 2$ such that Q is not a theta characteristic (i.e., $2Q \not\sim K_C$). Then $d(Q) + \mathbb{N}_0^l \subseteq \text{Sep}(C)$.*

§2.2. Special type and separating gonality. Let C be a separating $(M - 2)$ -curve of genus g , and let Teich_g^{M-2} denote the Teichmüller space of such curves. Denote the connected components of $\mathbb{R}C$ by $C_1, \dots, C_{g-1}$.

It follows from the result of Gabard [Gab06] that the separating gonality of C is either $g - 1$ or g . This gives the following partition of the Teichmüller space:

$$(\dagger) \quad \text{Teich}_g^{M-2} = \text{Teich}_{g,g}^{M-2} \cup \text{Teich}_{g,g-1}^{M-2}.$$

Coppens introduced another partition of Teich_g^{M-2} , defined in terms of curves of *special type*.

Definition 2.6. We say that a separating $(M - 2)$ -curve C of genus g is of *special type* if there exists a *special* connected component $C' \subset \mathbb{R}C$, that is, a connected component with the following property: for every degree- g morphism $f: C \rightarrow \mathbb{P}^1$ whose restriction to every connected component $C'' \neq C'$ has odd degree, we have $f(C') = \mathbb{R}\mathbb{P}^1$.

Otherwise, C is said to be *not of special type*.

For a real morphism $f: C \rightarrow \mathbb{P}^1$, we define its *topological degree vector* by

$$\delta(f) = (\delta_1(f), \dots, \delta_{g-1}(f)) \in \mathbb{Z}^l,$$

where $\delta_i(f)$ is the absolute value of the topological degree of the map

$$f|_{C_i}: C_i \approx S^1 \longrightarrow \mathbb{R}\mathbb{P}^1 \approx S^1.$$

Thus $\delta_i(f) \in \mathbb{Z}_{\geq 0}$ and does not depend on any choice of orientations. If f is separating, then $\delta_i(f)$ is simply the degree of the covering $C_i \rightarrow \mathbb{R}\mathbb{P}^1$, so that $\delta(f) = d(f)$.

Remark 2.7. We use this slightly non-standard notion of topological degree for the following reason. A morphism $C \rightarrow \mathbb{P}^1$ associated with a complete pencil $|D|$ is determined only up to postcomposition by a real automorphism of $\mathbb{P}^1$, or equivalently, up to a change of real basis in $H^0(C, \mathcal{O}_C(D))$. Since such an automorphism may reverse the orientation of $\mathbb{R}\mathbb{P}^1$, the usual oriented topological degree vector of $f|_{|D|}|_{\mathbb{R}C}$ is determined only up to simultaneously changing the signs of all its components. Taking absolute values removes this ambiguity.

Definition 2.6 can be reformulated as follows: a component $C' \subset \mathbb{R}C$ is special if every morphism $f: C \rightarrow \mathbb{P}^1$ of degree g whose restriction to every other component of $\mathbb{R}C$ has odd topological degree is separating. Such a morphism has topological degree $(2, 1, \dots, 1)$ after placing C' first. Thus C is not of special type if, for every component C' , there exists a nonseparating morphism of degree g with odd degree on all other real components and with $f(C') \neq \mathbb{R}\mathbb{P}^1$. Its topological degree is $(0, 1, \dots, 1)$ after the same ordering.

Thus we obtain a partition of the Teichmüller space “by type”:

$$(\ddagger) \quad \text{Teich}_g^{M-2} = \text{Teich}_{g,s}^{M-2} \cup \text{Teich}_{g,ns}^{M-2},$$

where $\text{Teich}_{g,s}^{M-2}$ and $\text{Teich}_{g,ns}^{M-2}$ denote the subspaces of curves of special and non-special type, respectively.

In [Cop14] Coppens studies the interaction between the partitions $(\dagger)$ and $(\ddagger)$. In particular, Coppens proves the following statements.

- (1) A separating $(M - 2)$ -curve that is not of special type has separating gonality $g - 1$.
- (2) There exist curves of special type with separating gonality $g - 1$, as well as curves of special type with separating gonality g .
- (3) If C is of special type and $\text{sepgon}(C) = g - 1$, then every separating pencil of degree $g - 1$ on C is a theta characteristic.
- (4) The locus $\text{Teich}_{g,ns}^{M-2}$ is open and $\overline{\text{Teich}_{g,ns}^{M-2}} = \text{Teich}_{g,g-1}^{M-2}$. Consequently, $\text{Teich}_{g,g}^{M-2} \subset \text{Teich}_{g,s}^{M-2}$.

The separating gonality and special type of low-genus separating $(M - 2)$ -curves are discussed in Appendix B.

§2.3. Brill–Noether theory. Let C be a smooth projective irreducible curve. For integers d and $r \geq 0$, we denote the corresponding Brill–Noether locus by

$$W_d^r(C) := \{[D] \in \text{Pic}^d(C) \mid h^0(D) \geq r + 1\}.$$

If the curve C is real, this subspace is invariant under the induced antiholomorphic involution and thus it is defined over $\mathbb{R}$. We denote its set of real points by $\mathbb{R}W_d^r(C)$.

We will need the following Brill–Noether-theoretic result.

Theorem 2.8 (Martens [Mar67]). *Let C be a smooth projective curve of genus g . Suppose that $0 < 2r \leq d < g$. Then $\dim W_d^r(C) \leq d - 2r$. Moreover, equality holds if and only if C is hyperelliptic.*

We will use the following description of the locus of base-point-free pencils and the resulting consequence of Martens’ bound. We believe that this statement is known (for example, part of it is stated in [CH11, p. 16]), but for the convenience of the reader, we give a complete proof here.

Proposition 2.9. *Let C be a smooth projective curve and let $d \geq 1$. Define*

$$W_d^{1,\circ}(C) = \{[D] \in W_d^1(C) \mid h^0(D) = 2, |D| \text{ is base-point-free}\}.$$

Then $W_d^{1,\circ}(C)$ is Zariski open in $W_d^1(C)$. If, in addition, C is non-hyperelliptic of genus g and $3 \leq d \leq g$, then

$$\dim(W_d^1(C) \setminus W_d^{1,\circ}(C)) \leq d - 3.$$

Proof. We claim that

$$W_d^1(C) \setminus W_d^{1,\circ}(C) = W_{d-1}^1(C) + C,$$

where the sum is taken in $\text{Pic}^d(C)$. Indeed, if $[D] \notin W_d^{1,\circ}(C)$, then either $|D|$ has a base point p , in which case $h^0(D - p) = h^0(D) \geq 2$, or $h^0(D) \geq 3$, in which case $h^0(D - p) \geq 2$ for every $p \in C$. Thus $[D] \in W_{d-1}^1(C) + C$.

Conversely, suppose that $D \sim E + p$ for some $[E] \in W_{d-1}^1(C)$ and $p \in C$. If $h^0(D) = h^0(E)$, then p is a base point of $|D|$; otherwise $h^0(D) > h^0(E) \geq 2$, and hence $h^0(D) \geq 3$. In either case, $[D] \notin W_d^{1,\circ}(C)$.

The locus $W_{d-1}^1(C) + C$ is Zariski closed in $\text{Pic}^d(C)$. Therefore $W_d^{1,\circ}(C)$ is Zariski open in $W_d^1(C)$.

Finally, suppose that C is non-hyperelliptic and $3 \leq d \leq g$. Martens’ theorem applied to $W_{d-1}^1(C)$ gives

$$\dim W_{d-1}^1(C) \leq d - 4.$$

Consequently,

$$\dim(W_d^1(C) \setminus W_d^{1,\circ}(C)) \leq \dim W_{d-1}^1(C) + 1 \leq d - 3.$$

□

Remark 2.10. As observed in [CH11, p. 15] and in [Cop14], we shall repeatedly use the standard fact that the topological degree vector

$$[D] \mapsto \delta(f_{|D|})$$

is locally constant on the locus $\mathbb{R}W_d^{1,\circ}(C)$ of complete base-point-free pencils. It can be seen directly using the Poincaré line bundle and the homotopy invariance of topological degree.

For completeness, we record here the following general openness property of separating pencils, although it will not be used later.

Let C be a separating real curve with connected components $C_1, \dots, C_l$. Consider the subspace $\mathcal{S}_d^1(C) \subset \mathbb{R}W_d^1(C)$ of separating pencils:

$$\mathcal{S}_d^1(C) := \left\{ [D] \in \mathbb{R}W_d^{1,\circ}(C) \mid f_{|D|}: C \rightarrow \mathbb{P}^1 \text{ is separating} \right\}.$$

Lemma 2.11. *The subset $\mathcal{S}_d^1(C)$ is open in $\mathbb{R}W_d^1(C)$.*

Proof. Since $\mathbb{R}W_d^{1,\circ}(C)$ is open in $\mathbb{R}W_d^1(C)$ and contains $\mathcal{S}_d^1(C)$, it is enough to prove that $\mathcal{S}_d^1(C)$ is open in $\mathbb{R}W_d^{1,\circ}(C)$. The morphism $f_{|D|}$ is separating if and only if

$$\sum_{i=1}^l \delta_i(f_{|D|}) = d.$$

Topological degree is locally constant on $\mathbb{R}W_d^{1,\circ}(C)$. Therefore, the above equality defines an open subset of $\mathbb{R}W_d^{1,\circ}(C)$. $\square$

§3. Proofs of the main results

§3.1. Curves of special type.

Theorem 3.1. *Let C be a separating $(M-2)$ -curve of genus $g \geq 4$.*

- (a) *Assume that $\text{sepgon}(C) = g$ and let f be a separating morphism of degree g . Number the connected components of $\mathbb{R}C$ so that $d_1(f) = 2$. Then*

$$(2, 1, \dots, 1) + \mathbb{N}_0^{g-1} \subseteq \text{Sep}(C).$$

- (b) *Assume that $\text{sepgon}(C) = g-1$ and C is of special type with a special component C_1 . Then*

$$\{(1, \dots, 1)\} \cup ((2, 1, \dots, 1) + \mathbb{N}_0^{g-1}) \subseteq \text{Sep}(C).$$

Proof. (a) Let P be a real fiber of f . By Riemann–Roch and the fact that $h^0(P) \geq 2$, we have $h^0(K_C - P) \geq 1$. Choose a real effective divisor $E \in |K_C - P|$. The canonical class has even degree on every component of $\mathbb{R}C$, whereas P has even degree on C_1 and odd degree on every other component. Hence E has odd degree on $C_2, \dots, C_{g-1}$. Since $\deg E = g-2$, it consists of exactly one point on each of these components and no other points. Proposition 2.1 therefore gives

$$h^1(P) = h^0(K_C - P) = h^0(E) = 1.$$

Thus the claim follows from part (a) of Theorem 2.5.

(b) Since the number of connected components of the real locus of a separating hyperelliptic curve of genus g can be 1, 2, or $g+1$ (see [GH81, Prop. 6.3]), a real $(M-2)$ -curve cannot be hyperelliptic for $g \geq 4$.

Since $\text{sepgon}(C) = g-1$, we first have

$$(1, \dots, 1) \in \text{Sep}(C).$$

It remains to prove that $(2, 1, \dots, 1) + \mathbb{N}_0^{g-1} \subseteq \text{Sep}(C)$. We first show that there exists an effective divisor

$$Q = q_2 + \dots + q_{g-1}, \quad q_i \in C_i,$$

such that the complete linear system $|K_C - Q|$ is base-point-free. Identifying $C_2 \times \dots \times C_{g-1}$ with the corresponding subspace of $\mathbb{R} \text{Sym}^{g-2}(C)$, consider the restriction of the Abel–Jacobi map

$$\varphi: C_2 \times \dots \times C_{g-1} \longrightarrow \mathbb{R} \text{Pic}^{g-2}(C), \quad (q_2, \dots, q_{g-1}) \longmapsto [Q],$$

where $Q = q_2 + \cdots + q_{g-1}$. Consider also the isomorphism

$$\iota_K: \mathbb{R} \operatorname{Pic}^{g-2}(C) \longrightarrow \mathbb{R} \operatorname{Pic}^g(C), \quad [D] \longmapsto [K_C - D].$$

Set $\Phi = \iota_K \circ \varphi$ and $\Lambda = \operatorname{Im}(\Phi)$. For every Q as above, by Proposition 2.1 we have $h^0(Q) = 1$, and hence by Riemann–Roch theorem we have $h^0(K_C - Q) = 2$. Therefore, $\Lambda \subset \mathbb{R}(W_g^1(C) \setminus W_g^2(C))$.

The differential of the Abel–Jacobi map at Q has rank $g - h^0(K_C - Q) = g - 2$ and thus it is injective. Its restriction to the real tangent space of $C_2 \times \cdots \times C_{g-1}$ is therefore also injective. Since ι_K is an isomorphism, it follows that $d_Q \Phi$ is injective and thus Φ is a real-analytic immersion.

The map Φ is also injective. Indeed, if $\Phi(Q) = \Phi(Q')$, then $Q \sim Q'$. Since $h^0(Q) = 1$, the complete linear system $|Q|$ consists of the single divisor Q , and hence $Q = Q'$. Finally, the domain of Φ is compact, so Φ is a proper injective immersion and therefore an embedding. Consequently, Λ is a smooth real submanifold of $\mathbb{R} \operatorname{Pic}^g(C)$ of dimension $\dim_{\mathbb{R}} \Lambda = g - 2$.

On the other hand, Proposition 2.9 gives

$$\dim_{\mathbb{R}}(\mathbb{R}W_g^1(C) \setminus \mathbb{R}W_g^{1,\circ}(C)) \leq g - 3.$$

Therefore, Λ is not contained in this locus.

Choose Q such that $[K_C - Q] \in \mathbb{R}W_g^{1,\circ}(C)$, and let $f: C \rightarrow \mathbb{P}^1$ be the corresponding morphism. The canonical class has even degree on every component of $\mathbb{R}C$. Therefore f has odd topological degree on $C_2, \dots, C_{g-1}$. Since C_1 is special, f is separating with $d(f) = (2, 1, \dots, 1)$.

Let D be a real fiber of f . Since $D \sim K_C - Q$, we have $h^1(D) = h^0(Q) = 1$ by Proposition 2.1. Part (a) of Theorem 2.5 now gives

$$(2, 1, \dots, 1) + \mathbb{N}_0^{g-1} \subseteq \operatorname{Sep}(C),$$

as required. $\square$

§3.2. Curves not of special type. We now turn to curves which are not of special type. One of the main results of [Cop14] states that for an $(M - 2)$ -curve C of special type with $\operatorname{sepgon}(C) = g - 1$, every separating pencil of degree $g - 1$ is a theta characteristic. We prove the converse: if C is not of special type, then it admits a separating pencil of degree $g - 1$ which is not a theta characteristic. This is precisely what is needed to apply Theorem 2.5(b).

To construct a separating pencil that is not a theta characteristic, we use Lemma 2.4, which translates the construction of separating divisors into a problem in linear algebra and convex geometry (cf. [Mag26, proof of Theorem 1]). Thus we begin with the technical convex-geometric lemma, whose proof is deferred to Appendix A.

Lemma 3.2. *Let V be a real vector space of dimension $n \geq 3$, and let*

$$u_1, \dots, u_n, w_1, \dots, w_n \in V.$$

Suppose that these vectors satisfy the following conditions:

- (1) $\sum_{i=1}^n u_i = 0$ and $\sum_{i=1}^n w_i = 0$;
- (2) $\dim \operatorname{span}(u_1, \dots, u_n) = \dim \operatorname{span}(w_1, \dots, w_n) = n - 1$;
- (3) $\operatorname{span}(u_1, \dots, u_n, w_1, \dots, w_n) = V$;
- (4) *every non-zero relation of the form*

$$\sum_{i=1}^n a_i u_i + \sum_{i=1}^n b_i w_i = 0, \quad a_i, b_i \geq 0,$$

- contains at least one of u_i, w_i in its support¹ for every $i = 1, \dots, n$;
 (5) if the support of such a relation contains exactly one of u_i, w_i for every $i = 1, \dots, n$, then it is either $\{u_1, \dots, u_n\}$ or $\{w_1, \dots, w_n\}$.

Then there exist an index j , vectors $z_i \in \{u_i, w_i\}$ for $i \neq j$, and positive real numbers

$$\alpha_i > 0 \quad (i \neq j), \quad \beta > 0, \quad \gamma > 0,$$

such that

$$\beta u_j + \gamma w_j + \sum_{i \neq j} \alpha_i z_i = 0.$$

Lemma 3.3. Assume that $\text{sepgon}(C) = g - 1$ and that every separating pencil of degree $g - 1$ on C is a theta characteristic. Then, after a suitable ordering of the connected components of $\mathbb{R}C$, there exist points $q_i \in C_i$, $i = 2, \dots, g - 1$, such that $|K_C - q_2 - \dots - q_{g-1}|$ is a separating pencil of topological degree $(2, 1, \dots, 1)$.

Proof. Put $n := g - 1 = b_0(\mathbb{R}C)$. Since $\text{sepgon}(C) = n$, there exists a separating morphism $f: C \rightarrow \mathbb{P}^1$ of degree n . Let L be its real fiber. By Lemma 2.2 $h^0(L) = 2$ and by assumption, $|L|$ is a theta characteristic.

Consider two divisors $F, G \in |L|$, put $F = p_1 + \dots + p_n$ and $G = q_1 + \dots + q_n$ where $p_i, q_i \in C_i$, $\text{Supp}(F) \cap \text{Supp}(G) = \emptyset$ with the following additional property. For every choice $r_i \in \{p_i, q_i\}$, $i = 1, \dots, n$, consider the mixed divisor $R = r_1 + \dots + r_n$. We claim that we may choose G sufficiently close to F so that the following holds:

$$(1) \quad 2R \sim K_C \implies [R] = [F].$$

Indeed, theta characteristics form a finite subset of $\text{Pic}^{g-1}(C)$. Hence there exists a neighborhood U of $[L]$ which contains no other theta characteristic. As G tends to F , all the finitely many mixed divisors R tend to F . Thus, after choosing G sufficiently close to F , the class of every mixed divisor belongs to U , and (1) follows.

Fix one of the two complex orientations of $\mathbb{R}C$, and let $\omega_1, \dots, \omega_g$ be a real basis of the space of holomorphic differentials on C . Applying Lemma 2.4 to the separating divisor F , we obtain tangent vectors $v_i \in T_{p_i}C_i$, $i = 1, \dots, n$, positive with respect to this orientation, such that

$$\sum_{i=1}^n \omega_k(v_i) = 0, \quad k = 1, \dots, g.$$

Define the vectors $u_i := (\omega_1(v_i), \dots, \omega_g(v_i)) \in \mathbb{R}^g$. Similarly, for G we obtain positive² tangent vectors $v'_i \in T_{q_i}C_i$, $i = 1, \dots, n$, with

$$\sum_{i=1}^n \omega_k(v'_i) = 0, \quad k = 1, \dots, g.$$

Define also the vectors $w_i = (\omega_1(v'_i), \dots, \omega_g(v'_i))$. Set

$$V = \text{span}(u_1, \dots, u_n, w_1, \dots, w_n) \subset \mathbb{R}^g.$$

We now verify that $u_1, \dots, u_n, w_1, \dots, w_n$ satisfy the assumptions of Lemma 3.2.

- (1) Follows from Lemma 2.4.
- (2) We have $\dim \text{span}(u_1, \dots, u_n) = g - h^0(K_C - F) = g - h^0(F) = g - 2 = n - 1$ and similarly for w_i .
- (3) $\dim V = g - h^0(K_C - F - G) = g - h^0(\mathcal{O}_C) = g - 1 = n$.

¹The support of a linear combination $\sum_i \lambda_i v_i$ is the set of those v_i for which $\lambda_i \neq 0$.

²Without loss of generality we may assume that v_i and v'_i are positive with respect to the same complex orientation.

- (4) Now consider a non-zero relation with nonnegative coefficients

$$\sum_{i=1}^n a_i u_i + \sum_{i=1}^n b_i w_i = 0, \quad a_i, b_i \geq 0.$$

Let S be its support, and let D_S be the divisor consisting of the points p_i for which $a_i > 0$ and the points q_i for which $b_i > 0$. After replacing the tangent vectors occurring in D_S by $a_i v_i$ and $b_i v'_i$, respectively, we obtain tangent vectors at the points of D_S satisfying the assumptions of Lemma 2.4 for D_S . Therefore D_S is separating. A separating divisor has at least one point on every connected component of $\mathbb{R}C$. Hence S contains at least one of u_i, w_i for every i .

- (5) Suppose that in the notation of the previous item the support S contains exactly one of u_i, w_i for every i . The corresponding divisor is of the form

$$R = r_1 + \cdots + r_n, \quad r_i \in \{p_i, q_i\}.$$

It follows from the above that R is a separating divisor of degree $n = g - 1$. By our assumption, its divisor class is a theta characteristic. Property (1) therefore gives $[R] = [F] = [L]$, and hence $R \in |L|$. Since $h^0(L) = 2$, the divisors in $|L|$ are precisely the fibers of f . If the choice of the points r_i is mixed, then R contains at least one point of F and at least one point of G . These points have different images under f , because F and G are distinct fibers. Hence a mixed divisor cannot belong to $|L|$. It follows that the only possible supports containing exactly one element from each pair are

$$\{u_1, \dots, u_n\} \quad \text{and} \quad \{w_1, \dots, w_n\}.$$

Thus all the hypotheses of Lemma 3.2 are satisfied. Combining Lemma 3.2 with Lemma 2.4, we obtain an index j and a separating divisor

$$P = p_j + q_j + \sum_{i \neq j} r_i, \quad r_i \in \{p_i, q_i\}.$$

For $i \neq j$, let r'_i be the point of the pair $\{p_i, q_i\}$ different from r_i , and put

$$Q = \sum_{i \neq j} r'_i.$$

Then $P + Q = F + G$. Since $F, G \in |L|$ and L is a theta characteristic,

$$P + Q \sim 2L \sim K_C,$$

so $P \sim K_C - Q$. The divisor Q consists of $g - 2$ points on distinct components of $\mathbb{R}C$; hence Proposition 2.1 gives $h^0(Q) = 1$. Riemann–Roch therefore yields

$$h^0(K_C - Q) = h^0(Q) + 1 = 2.$$

Thus $|K_C - Q|$ is a separating pencil. Renumbering the components so that $j = 1$ gives topological degree $(2, 1, \dots, 1)$. □

Lemma 3.4. *Suppose that $\text{sepgon}(C) = g - 1$, where $g \geq 5$, and that C is not of special type. Then there exists a separating pencil of degree $g - 1$ that is not a theta characteristic.*

Proof. Assume, to the contrary, that every separating pencil of degree $g - 1$ is a theta characteristic. After ordering the components as in Lemma 3.3, set

$$\mathcal{T} = C_2 \times \cdots \times C_{g-1},$$

and identify a point $(q_2, \dots, q_{g-1}) \in \mathcal{T}$ with the divisor $Q = q_2 + \dots + q_{g-1}$. By Proposition 2.1 and Riemann–Roch, we have $h^0(K_C - Q) = 2$ for every $Q \in \mathcal{T}$.

Consider the locus of the divisors $Q \in \mathcal{T}$ so that $|K_C - Q|$ has a base point on the component C_1 :

$$\mathcal{B}_1 = \{Q \in \mathcal{T} \mid |K_C - Q| \text{ has a base point on } C_1\}.$$

Let $Q \in \mathcal{B}_1$, let $p \in C_1$ be a base point of $|K_C - Q|$. Then $h^0(K_C - Q - p) = h^0(K_C - Q) = 2$. By the Riemann–Roch theorem $h^0(Q + p) = 2$.

Removing any point from $Q + p$ leaves $g - 2$ points on distinct components, so Proposition 2.1 shows that $|Q + p|$ is base-point-free. Therefore by Lemma 2.3 $|Q + p|$ is a separating pencil of degree $g - 1$. Then by our assumption, $Q + p \in |E|$ where $|E|$ is one of the finitely many theta characteristics on C . Since $|E|$ is a separating pencil of degree $g - 1$, we have $d_i(f|_E) = 1$ for every i . Hence each restriction

$$f|_E|_{C_i}: C_i \longrightarrow \mathbb{RP}^1$$

is a homeomorphism. Denote its inverse by $q_i: \mathbb{RP}^1 \rightarrow C_i$. Thus, for every $t \in \mathbb{RP}^1$,

$$f|_E^{-1}(t) = q_1(t) + \dots + q_{g-1}(t).$$

Consequently, the map

$$\gamma|_E: \mathbb{RP}^1 \longrightarrow \mathcal{T}, \quad t \longmapsto (q_2(t), \dots, q_{g-1}(t)),$$

is an embedding, and its image is a circle in $\mathcal{T}$. Since $Q + p \in |E|$, we have $Q + p = f|_E^{-1}(t_0)$ for some $t_0 \in \mathbb{RP}^1$, and therefore $Q = \gamma|_E(t_0)$.

Since there are only finitely many theta characteristics on C , it follows that $\mathcal{B}_1$ is contained in a finite union of circles. As $\dim_{\mathbb{R}} \mathcal{T} = g - 2 \geq 3$, such a subset cannot disconnect $\mathcal{T}$. Hence $\mathcal{T} \setminus \mathcal{B}_1$ is connected.

By definition, for $Q \in \mathcal{T} \setminus \mathcal{B}_1$, the pencil $|K_C - Q|$ has no base points on C_1 . Hence its restriction to C_1 defines a continuous map

$$\varphi_Q: C_1 \approx S^1 \longrightarrow S^1 \approx \mathbb{RP}^1.$$

Although its topological degree is defined only up to sign, its vanishing is well defined. By the same local-trivialization argument as in Remark 2.10, the condition $\deg(\varphi_Q) = 0$ is locally constant on $\mathcal{T} \setminus \mathcal{B}_1$. Since this space is connected, the condition must either hold for every $Q \in \mathcal{T} \setminus \mathcal{B}_1$ or for none of them. It therefore suffices to find points $Q_+, Q_- \in \mathcal{T} \setminus \mathcal{B}_1$ such that φ_{Q_+} has non-zero topological degree, whereas $\deg(\varphi_{Q_-}) = 0$.

By Lemma 3.3, there exists $Q_+ \in \mathcal{T} \setminus \mathcal{B}_1$ for which $|K_C - Q_+|$ is separating of topological degree $(2, 1, \dots, 1)$. Thus $\deg(\varphi_{Q_+}) = 2$.

On the other hand, since C_1 is not a special component, there exists a real morphism $h: C \rightarrow \mathbb{P}^1$ of degree g such that $h(C_1) \subsetneq \mathbb{RP}^1$ and $\delta_i(h)$ is odd for $i = 2, \dots, g - 1$. Let P be a real fiber of h . Riemann–Roch gives $h^0(K_C - P) \geq 1$. Choose a real effective divisor $Q_- \sim K_C - P$. Its degree is $g - 2$, and parity on the real components forces

$$Q_- = q_2 + \dots + q_{g-1}, \quad q_i \in C_i.$$

Thus $Q_- \in \mathcal{T}$. Proposition 2.1 gives $h^0(Q_-) = 1$, and Riemann–Roch then shows that $h^0(P) = 2$. Therefore $|P| = |K_C - Q_-|$ is precisely the pencil defining h . This pencil is base-point-free, and the proper inclusion $h(C_1) \subsetneq \mathbb{RP}^1$ implies that $\deg(\varphi_{Q_-}) = 0$. $\square$

Remark 3.5. The proof of Lemma 3.4 is inspired by Coppens’s proof of Theorem 4.4 in [Cop14], where topological degrees are compared along a path of pencils parametrized by $C_2 \times \dots \times C_{g-1}$. Here we refine this idea by analyzing the locus where the pencil has a base point on C_1 and showing that its complement is connected.

Combining Lemma 3.4 with [Cop14, Proposition 4.1] we obtain another interesting characterization of curves of special type with $\text{sepgon}(C) = g - 1$.

Corollary 3.6. *Let C be a separating $(M - 2)$ -curve of genus $g \geq 5$ with $\text{sepgon}(C) = g - 1$. Then C is of special type if and only if every separating pencil of degree $g - 1$ on C is a theta characteristic.*

Theorem 3.7. *Let C be a separating $(M - 2)$ -curve of genus g . Assume that $g \geq 5$ and C is not of special type. Then*

$$\text{Sep}(C) = \mathbb{N}^{g-1}.$$

Proof. We apply part (b) of Theorem 2.5. A real $(M - 2)$ -curve cannot be hyperelliptic for $g \geq 4$ (see the beginning of the proof of part b) of Theorem 3.1).

By Lemma 3.4, there exists a separating pencil $|Q|$ of degree $g - 1$ such that $2Q \not\sim K_C$. Its topological degree is $(1, \dots, 1)$, and Lemma 2.2 gives $h^0(Q) = 2$. Part (b) of Theorem 2.5 therefore yields

$$(1, \dots, 1) + \mathbb{N}_0^{g-1} = \mathbb{N}^{g-1} \subseteq \text{Sep}(C),$$

hence $\text{Sep}(C) = \mathbb{N}^{g-1}$. □

As mentioned in §2.2, Coppens [Cop14, Lemma 2.6 and Corollary 4.2] showed that $\text{Teich}_{g,ns}^{M-2}$ is a non-empty open subset of Teich_g^{M-2} . Combining this result with Theorem 3.7, we obtain the following corollary.

Corollary 3.8. *Let $g \geq 5$. On a non-empty open subset of Teich_g^{M-2} , the separating semigroup equals $\mathbb{N}^{g-1}$.*

Appendix A. A convex-geometric lemma

Lemma 3.2. *Let V be a real vector space of dimension $n \geq 3$, and let*

$$u_1, \dots, u_n, w_1, \dots, w_n \in V.$$

Suppose that these vectors satisfy the following conditions:

- (1) $\sum_{i=1}^n u_i = 0$ and $\sum_{i=1}^n w_i = 0$;
- (2) $\dim \text{span}(u_1, \dots, u_n) = \dim \text{span}(w_1, \dots, w_n) = n - 1$;
- (3) $\text{span}(u_1, \dots, u_n, w_1, \dots, w_n) = V$;
- (4) *every non-zero relation of the form*

$$\sum_{i=1}^n a_i u_i + \sum_{i=1}^n b_i w_i = 0, \quad a_i, b_i \geq 0,$$

contains at least one of u_i, w_i in its support³ for every $i = 1, \dots, n$;

- (5) *if the support of such a relation contains exactly one of u_i, w_i for every $i = 1, \dots, n$, then it is either $\{u_1, \dots, u_n\}$ or $\{w_1, \dots, w_n\}$.*

Then there exist an index j , vectors $z_i \in \{u_i, w_i\}$ for $i \neq j$, and positive real numbers

$$\alpha_i > 0 \quad (i \neq j), \quad \beta > 0, \quad \gamma > 0,$$

such that

$$\beta u_j + \gamma w_j + \sum_{i \neq j} \alpha_i z_i = 0.$$

³The support of a linear combination $\sum_i \lambda_i v_i$ is the set of those v_i for which $\lambda_i \neq 0$.

Proof. Let $A: \mathbb{R}^{2n} \rightarrow V$ be the linear map whose columns are $u_1, \dots, u_n, w_1, \dots, w_n$, i.e.

$$A(a_1, \dots, a_n, b_1, \dots, b_n) = \sum_{i=1}^n a_i u_i + \sum_{i=1}^n b_i w_i.$$

By assumption (3) we have

$$\text{rank}(A) = \dim \text{span}(u_1, \dots, u_n, w_1, \dots, w_n) = \dim V = n$$

and thus $\dim \text{Ker}(A) = n$. Consider a polyhedral cone

$$\mathcal{C} := \text{Ker}(A) \cap (\mathbb{R}_{\geq 0})^{2n}.$$

Since $\sum_{i=1}^n u_i = 0$ and $\sum_{i=1}^n w_i = 0$, we have

$$e_u = (1, \dots, 1, 0, \dots, 0) \in \mathcal{C} \text{ and } e_w = (0, \dots, 0, 1, \dots, 1) \in \mathcal{C}.$$

Therefore $e_u + e_w \in \mathcal{C}$ and hence $\mathcal{C}$ is full-dimensional in $\text{Ker}(A)$ (since it contains an interior point of $(\mathbb{R}_{\geq 0})^{2n}$).

We are interested in the set $e(\mathcal{C})$ of extreme rays of $\mathcal{C}$. First we claim that $\#e(\mathcal{C}) \geq n$. Indeed, consider a compact polytope

$$\mathcal{P} := \left\{ x \in \mathcal{C} \mid \sum_{i=1}^{2n} x_i = 1 \right\}.$$

Clearly, the vertices of $\mathcal{P}$ are in one-to-one correspondence with extreme rays of $\mathcal{C}$. On the other hand, $\dim \mathcal{P} = n - 1$, so $\mathcal{P}$ has at least n vertices.

Secondly, we claim that $e_u \cdot \mathbb{R}_{\geq 0} \in e(\mathcal{C})$ and $e_w \cdot \mathbb{R}_{\geq 0} \in e(\mathcal{C})$. Suppose, for a contradiction, that $e_u = x + y$. Since the last n coordinates of e_u vanish and all coordinates of x and y are nonnegative, the last n coordinates of both x and y vanish. Thus x and y determine nonnegative linear relations among $u_1, \dots, u_n$. By assumption (2), the space of all linear relations among $u_1, \dots, u_n$ is one-dimensional. Since

$$u_1 + \dots + u_n = 0,$$

this space is generated by $(1, \dots, 1)$. Hence both x and y are nonnegative multiples of e_u . This proves that $\mathbb{R}_{\geq 0} e_u$ is an extreme ray. The same argument applies to e_w .

Since $\#e(\mathcal{C}) \geq n \geq 3$, the cone $\mathcal{C}$ contains an extreme ray different from $e_u \cdot \mathbb{R}_{\geq 0}$ and $e_w \cdot \mathbb{R}_{\geq 0}$. Choose such a ray and denote it by $\rho \cdot \mathbb{R}_{\geq 0}$, where $\rho = (a_1, \dots, a_n, b_1, \dots, b_n)$. We now prove that the coordinates of ρ give the desired linear combination.

First let us prove that ρ has exactly $n + 1$ non-zero coordinates. Denote $I := \{i = 1, \dots, 2n \mid \rho_i \neq 0\}$. We next show that $\#I \leq n + 1$. For $v \in \mathbb{R}^{2n}$ denote by v_I the vector from $\mathbb{R}^{\#I}$ formed by the coordinates of v belonging to I . Let $A_I: \mathbb{R}^{\#I} \rightarrow V$ denote the restriction of A to the coordinates belonging to I . We claim that $\dim \text{Ker}(A_I) = 1$. Suppose that there exists a vector $\sigma_I \in \text{Ker}(A_I)$ which is not proportional to ρ_I . Since all coordinates of ρ_I are positive, for a sufficiently small $\varepsilon > 0$ both vectors $\rho_I + \varepsilon \cdot \sigma_I$ and $\rho_I - \varepsilon \cdot \sigma_I$ have non-negative coordinates. Extending them by zeros outside I , we obtain two nonproportional elements of $\mathcal{C}$ whose average is ρ . This contradicts the extremality of ρ . Thus $\dim \text{Ker}(A_I) = 1$, so $\text{rank}(A_I) = \#I - 1$. On the other hand, $\text{rank}(A_I) \leq \text{rank}(A)$, thus $\#I \leq n + 1$.

However, assumption (4) implies that I contains at least one element of each pair $\{u_i, w_i\}$, $i = 1, \dots, n$. Therefore, $\#I \geq n$.

If $\#I = n$, then I contains exactly one element of each pair. Assumption (5) then implies that $\rho = (a_1, \dots, a_n, 0, \dots, 0)$ or $\rho = (0, \dots, 0, b_1, \dots, b_n)$. By assumption (2), the corresponding positive relation is unique up to multiplication by a positive scalar. Thus $\rho \cdot \mathbb{R}_{\geq 0} = e_u \cdot \mathbb{R}_{\geq 0}$ or $\rho \cdot \mathbb{R}_{\geq 0} = e_w \cdot \mathbb{R}_{\geq 0}$, contrary to the choice of ρ .

Consequently, $\#I = n + 1$. Since I meets all n pairs $\{u_i, w_i\}$, it contains both u_j and w_j for exactly one index j , and exactly one of u_i, w_i for every $i \neq j$. The coefficients of all vectors in the support are strictly positive, which gives the required relation. $\square$

Appendix B. Separating gonality and special type in low genus

Let us describe the separating gonality and the special type of separating $(M - 2)$ -curves of low genus (all of them, except for the simple genus-3 case, are shown in Figure 1).

Recall that if C is a real algebraic curve on a real algebraic surface S , then a connected component $C' \subset \mathbb{R}C$ is called an *oval* if it bounds a disk in $\mathbb{R}S$.

Genus 3. Suppose at first that C is non-hyperelliptic. Then C is a *hyperbolic quartic*, i.e., a smooth real plane quartic whose real locus consists of two nested ovals. Denote the outer oval by C_1 and the inner one by C_2 .

Projection from a point $p \in C_2$ defines a separating morphism of degree 3 and degree partition $(2, 1)$. On the other hand, since a smooth plane quartic is non-hyperelliptic, it admits no morphism of degree 2. Consequently,

$$\text{sepgon}(C) = 3 = g.$$

We claim that the component C_1 is special. Indeed, let $f: C \rightarrow \mathbb{P}^1$ be a degree-3 morphism having odd degree on C_2 , and let D be the divisor of a real fiber of f . By Riemann–Roch we have $h^0(D) = h^0(K_C - D) + 1$, so $h^0(K_C - D) \geq 1$. Therefore there exists an effective real divisor E such that $K_C \sim D + E$. Since $\deg(K_C - D) = 1$, there exists a point $p \in \mathbb{R}C$ such that $D + p \sim K_C$. The canonical divisor has even degree on every connected component of $\mathbb{R}C$ (see e.g. [Mag26, Lemma 3] or [Hui03, Prop. 2.1]), whereas D has odd degree on C_2 . Hence $p \in C_2$. By the adjunction formula, the canonical linear system of a plane quartic is cut out by lines. Therefore there exists a real line ℓ such that $\ell \cap \mathbb{R}C = D + p$. Since p lies on the inner oval, the line ℓ intersects the outer oval C_1 in two real points. Hence $\deg(D|_{C_1}) = 2$ and C_1 is special.

Now suppose that C is a hyperelliptic separating curve of genus 3 with 2 connected components. The hyperelliptic projection $\pi: C \rightarrow \mathbb{P}^1$ defines a separating morphism of degree 2 with $d(\pi) = (1, 1)$ and hence

$$\text{sepgon}(C) = 2 = g - 1.$$

Both components of $\mathbb{R}C$ are special, since C admits no morphism of degree 3 to $\mathbb{P}^1$. Indeed, suppose that such a morphism $f: C \rightarrow \mathbb{P}^1$ exists, and let $D = (f)_0$. Since $h^0(D) \geq 2$, Riemann–Roch gives $h^0(K_C - D) \geq 1$, so D is special. By [Ore19, Lemma 3.1], the morphism f factors through the hyperelliptic projection: $f = f_1 \circ \pi$ for some morphism $f_1: \mathbb{P}^1 \rightarrow \mathbb{P}^1$. Consequently, $\deg f = 2 \deg f_1$, contradicting $\deg f = 3$.

The separating semigroups in both cases were computed in [Ore19].

Genus 4. If C is non-hyperelliptic, its canonical image lies on an irreducible real quadric $S \subset \mathbb{P}^3$.

We number the components of $\mathbb{R}C$ so that the component denoted by c_2 in [Ore25] is C_1 .

- Suppose that S is an ellipsoid. This case is shown in Figure 1(a). Then all three components of $\mathbb{R}C$ are ovals. Orevkov shows that every separating morphism has degree at least 4 and constructs one of degree partition $(2, 1, 1)$ in our numbering. Hence

$$\text{sepgon}(C) = 4 = g.$$

Since a non-special $(M - 2)$ -curve has separating gonality $g - 1$, the curve C is of special type.

- Suppose that S is a quadratic cone and C is a curve without ovals on S . This case is shown in Figure 1(b). We number the connected components of $\mathbb{R}C$ so that

$$C_1 = c_2, \quad C_2 = c_1, \quad C_3 = c_3$$

in the notation of [Ore25].

Let $A \subset S$ be a real generator of the cone. The pencil of plane sections through A defines a separating morphism of degree 3 realizing $(1, 1, 1) \in \text{Sep}(C)$; see [Ore25, Proposition 3.2]. Hence

$$\text{sepgon}(C) = 3 = g - 1.$$

We claim that C_1 is special. Indeed, let $f: C \rightarrow \mathbb{P}^1$ be a degree-4 morphism having odd degree on C_2 and C_3 , and let D be the divisor of an arbitrary real fiber of f . By Riemann–Roch, $h^0(D) = h^0(K_C - D) + 1$, so $h^0(K_C - D) \geq 1$. Since $\deg(K_C - D) = 2$, there exists an effective real divisor E of degree 2 such that $D + E \sim K_C$. The canonical divisor has even degree on every connected component of $\mathbb{R}C$, whereas D has odd degree on C_2 and C_3 . Therefore

$$E = p_2 + p_3, \quad p_2 \in C_2, \quad p_3 \in C_3.$$

The canonical linear system of C is cut out by planes in $\mathbb{P}^3$. Hence $D + p_2 + p_3$ is cut out on C by a real plane containing the line $L = \overline{p_2 p_3}$. Moreover, the pencil $|D|$ is obtained from the pencil of planes through L by removing the fixed points p_2 and p_3 . The line L cannot be a generator of the cone. Indeed, if $L \subset S$, then $L \cdot C = 3$. Thus the pencil of planes through L has a fixed divisor of degree 3 on C , and hence its moving part has degree $\deg K_C - 3 = 6 - 3 = 3$. This contradicts the fact that the pencil $|D|$ defines the degree-4 morphism f . Therefore $L \not\subset S$. Since S is a quadric and L already meets S at p_2 and p_3 , it follows that $L \cap C_1 = \emptyset$. The pencil of planes through such a line has topological degree $(2, 1, 1)$ in our numbering and hence C_1 is a special component.

- Suppose that S is a quadratic cone and C has two ovals. This case is shown in Figure 1(c). Orevkov proves that no separating morphism of degree 3 exists, while a separating morphism of topological degree $(2, 1, 1)$ does exist. Therefore

$$\text{sepgon}(C) = 4 = g,$$

and hence C is of special type by the results of Coppens [Cop14].

- Suppose that S is a hyperboloid and C is a curve without ovals on S . This case is shown in Figure 1(d). Let A and B be real lines in the two rulings of S . Then S is isomorphic to $A \times B \cong \mathbb{P}^1 \times \mathbb{P}^1$, and we denote the corresponding projections by

$$\pi_A: S \rightarrow A, \quad \pi_B: S \rightarrow B.$$

The canonical curve C has class $C \sim 3A + 3B \in \text{Pic}(S) \cong \mathbb{Z} \oplus \mathbb{Z}$, so the restrictions

$$\pi_A|_C: C \rightarrow A \simeq \mathbb{P}^1, \quad \pi_B|_C: C \rightarrow B \simeq \mathbb{P}^1$$

are morphisms of degree 3 and realize $(1, 1, 1) \in \text{Sep}(C)$; see [Ore25, Proposition 3.10]. Hence

$$\text{sepgon}(C) = 3 = g - 1.$$

Let $L_A = A|_C$ and $L_B = B|_C$. By the adjunction formula,

$$K_C \sim L_A + L_B.$$

Moreover, it is not hard to prove that $L_A \not\sim L_B$. Therefore the separating pencil $|L_A|$ is not a theta characteristic: otherwise $2L_A \sim K_C$ would imply $L_A \sim L_B$. Thus C

admits a separating pencil of degree $g-1$ which is not a theta characteristic. It follows from [Cop14] that C is not of special type.

- Suppose that C is a curve with two ovals on a hyperboloid $S \simeq \mathbb{P}^1 \times \mathbb{P}^1$. This case is shown in Figure 1(e). We keep the notation A , B , π_A , and π_B introduced above. In this case Orevkov proves that there is no separating morphism of degree 3, whereas a separating morphism of degree 4 does exist; see [Ore25, Propositions 3.11 and 3.12]. Consequently,

$$\text{sepgon}(C) = 4 = g.$$

Since every $(M-2)$ -curve which is not of special type has separating gonality $g-1$, it follows from [Cop14] that C is of special type.

Plane quintic. Let C be a plane separating $(M-2)$ -quintic (in this case $g=6$). This case is shown in Figure 1(f). As in [MO25, §4], we number the connected components of $\mathbb{R}C$ so that C_1 is the pseudoline. We claim that C_1 is special. Let $f: C \rightarrow \mathbb{P}^1$ be a morphism of degree 6 having odd degree on every oval of $\mathbb{R}C$, and let D be a real fiber of f . Since $\deg D = g$ and $h^0(D) \geq 2$, the Riemann–Roch theorem gives $h^0(K_C - D) \geq 1$. Thus there exists an effective real divisor E of degree 4 such that $D + E \sim K_C$.

The canonical linear system of a plane quintic is cut out by conics, so $D + E$ is the intersection divisor of C with a real conic. Since a canonical divisor has even degree on every real component, whereas D has odd degree on each of the four ovals, the divisor E also has odd degree on each oval. As $\deg E = 4$, it follows that E contains exactly one point on each oval and no point on the pseudoline.

Using the same topological argument as in [Vir84, §§3.2–3.3] (cf. [MO25, §4]), one shows that every such real conic meets the pseudoline C_1 in two real points. Since E has no points on C_1 , these two points belong to D . Therefore $f(C_1) = \mathbb{RP}^1$ and C_1 is a special component.

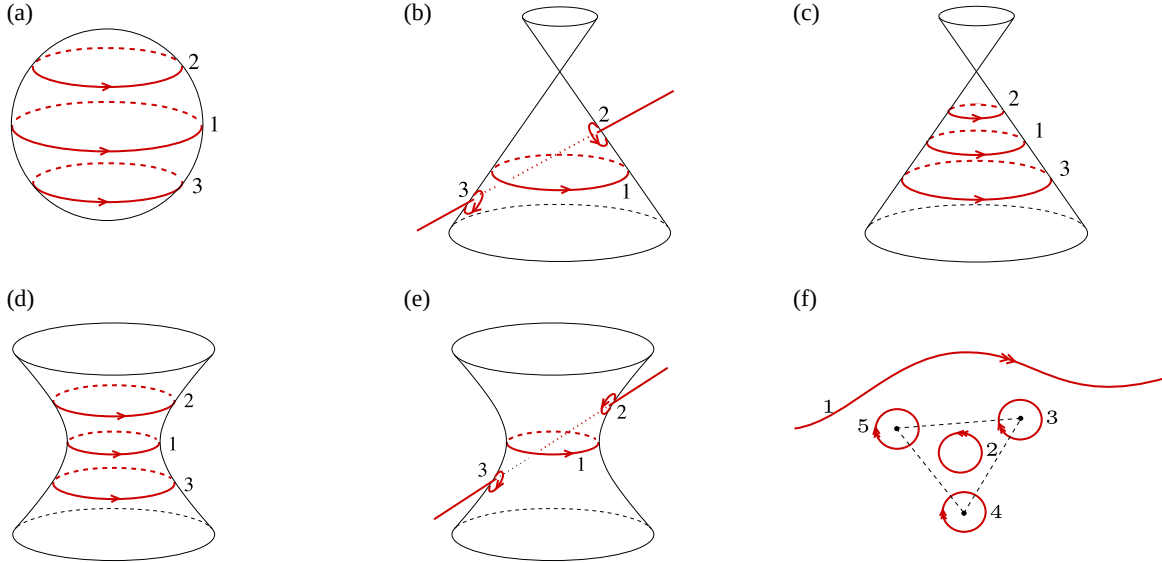


FIGURE 1. Separating $(M-2)$ -curves of genus 4 and a separating $(M-2)$ -quintic.

The above cases are summarized in Table 1.

g	Model	Ovals	$\text{sepgon}(C)$	Type	$\text{Sep}(C)$
3	Plane quartic	2	$3 = g$	Special	$(2, 1) + \mathbb{N}_0^2$
3	Hyperelliptic	–	$2 = g - 1$	Special	$\{(1, 1)\} \cup (\mathbb{N}_{\geq 2} \times \mathbb{N}_{\geq 2})$
4	Ellipsoid	3	$4 = g$	Special	$(2, 1, 1) + \mathbb{N}_0^3$
4	Quadratic cone	0	$3 = g - 1$	Special	$\{(1, 1, 1)\} \cup ((2, 1, 1) + \mathbb{N}_0^3)$
4	Quadratic cone	2	$4 = g$	Special	$(2, 1, 1) + \mathbb{N}_0^3$
4	Hyperboloid	0	$3 = g - 1$	Non-special	$\mathbb{N}^3$
4	Hyperboloid	2	$4 = g$	Special	$(2, 1, 1) + \mathbb{N}_0^3$
6	Plane quintic	4	$6 = g$	Special	$(2, 1, 1, 1, 1) + \mathbb{N}_0^5$

Table 1. Separating gonality, special type, and separating semigroups of some separating $(M - 2)$ -curves of low genus.

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